SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Bhuvaneshwari J	**Reg. No.:**	192572141
Date:	15/10/2026	**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>*.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Agile Sprint Planning & User Story Workshop
Project 29: Smart Construction Project Monitoring System (SCPMS)
Course: CSA10 / CSA1004 Software Engineering | CO1 | Assessment Tool 2 |
Student Name: Bhuvaneshwari J | Reg. No.: 192572141

1. Project Vision and Stakeholders

1.1 Problem Summary
Construction projects involve many contractors, materials, workers, schedules and budgets. When project information is late or inaccurate, delays and cost overruns follow. Managers need one system that shows the true state of every project in real time.

1.2 Vision Statement
For construction project managers who need accurate, real-time visibility of their projects, the Smart Construction Project Monitoring System is a web-based platform that tracks milestones, materials, workforce, expenses and risks in one place. Unlike spreadsheets and manual site reports, it detects delays and budget overruns early and generates progress reports automatically, so projects finish on time and within budget.

1.3 Project Objectives

#	**Objective**	**Measure of Success**
O1	Manage construction projects	A project and its team can be created in under 5 minutes
O2	Track project milestones	Milestone progress visible with at most 1 day of data lag
O3	Monitor material availability	Low-stock alert raised automatically at the reorder level
O4	Record workforce information	100% of site workers registered and assigned to a project
O5	Track project expenses	Actual vs budget shown per project and per category
O6	Identify project delays	Delayed milestones flagged automatically on the due date
O7	Manage risks and issues	Every risk has an owner, severity and status
O8	Generate progress reports	Report produced in under 30 seconds

1.4 Stakeholders

Stakeholder	Role / Interest	Key Needs
Project Manager (primary user)	Plans and controls the project	Real-time progress, delay alerts, reports
Site Engineer / Supervisor	Updates daily site data	Fast progress, material and attendance entry
Contractors / Subcontractors	Execute packages of work	Clear milestones, material availability
Store / Material Manager	Manages inventory	Stock records, low-stock alerts
Finance / Accounts Officer	Controls budget	Expense recording, budget vs actual
Client / Owner	Funds the project	Progress and cost transparency
Risk / Safety Officer	Manages risks and incidents	Risk register, issue tracking
System Administrator	Maintains the system	User and role management, security
Development Team, Scrum Master, Product Owner	Build and deliver the product	Clear backlog, realistic sprint plan
Regulatory Authorities	Compliance	Accurate records and reports

2. Epics and User Stories

2.1 Epics

Epic	Name	Description
E1	Project and User Management	Secure access, roles, and creation of projects
E2	Milestone and Schedule Tracking	Plan milestones, update progress, detect delays
E3	Material Management	Stock records, usage and shortage alerts
E4	Workforce Management	Worker records, assignment and attendance
E5	Expense and Budget Control	Expense entry and budget vs actual analysis
E6	Risk and Issue Management	Risk register and issue tracking
E7	Dashboard and Reporting	Real-time dashboard and progress reports

2.2 User Stories
Format: ***As a <user>, I want <feature>, so that <benefit>.***

ID	Epic	User Story
US-01	E1	As an administrator, I want to register users with roles (Manager, Engineer, Finance, Store), so that each person only accesses what they are allowed to.
US-02	E1	As a project manager, I want to create a construction project with name, location, budget, start and end dates, so that all activities are tracked under one project.
US-03	E2	As a project manager, I want to define milestones with planned start and end dates, so that I can monitor the schedule.
US-04	E2	As a site engineer, I want to update the completion percentage of a milestone, so that managers see the current progress.
US-05	E2	As a project manager, I want automatic delay alerts when a milestone is behind schedule, so that I can take corrective action early.
US-06	E3	As a store manager, I want to record material stock (item, quantity, unit, supplier), so that material availability is always known.
US-07	E3	As a store manager, I want a low-stock alert when quantity falls below the reorder level, so that work is not stopped by shortages.
US-08	E3	As a site engineer, I want to record material issued to a milestone, so that usage is traceable and stock updates automatically.
US-09	E4	As a project manager, I want to register workers (name, skill, contractor) and assign them to a project, so that workforce information is complete.
US-10	E4	As a site supervisor, I want to record daily attendance, so that labour hours and productivity can be monitored.
US-11	E5	As a finance officer, I want to record project expenses by category, so that spending is tracked accurately.
US-12	E5	As a project manager, I want to compare actual expenses with the budget, so that I can control cost overruns.
US-13	E6	As a risk officer, I want to log risks and issues with severity and owner, so that they are managed and not forgotten.
US-14	E6	As a project manager, I want to update risk status and mitigation actions, so that I can see which risks remain open.
US-15	E7	As a project manager, I want a dashboard of progress, delays, material, cost and risks, so that I get real-time project visibility.
US-16	E7	As a project manager, I want to generate and download progress reports, so that I can share status with the client and management.

3. Acceptance Criteria (Given-When-Then)

3.1 Sprint 1 Stories (detailed)

US-01 User registration and roles

- Given I am logged in as an administrator, when I submit a new user with a valid name, email, password and role, then the user is saved and can log in within 1 minute.
- Given an email already exists, when I register it again, then the system rejects it with the message "Email already registered".
- Given I am logged in as a Store user, when I open the Expenses page, then access is denied (HTTP 403).

US-02 Create project

- Given I am a project manager, when I enter name, location, budget (> 0), start date and end date and click Save, then the project appears in my project list.
- Given the end date is earlier than the start date, when I save, then the system shows a validation error and does not save.
- Given a project exists, when I open it, then all saved details display correctly.

US-03 Define milestones

- Given a project exists, when I add a milestone with name, planned start and planned end within the project dates, then it is saved with 0% progress.
- Given a planned date falls outside the project dates, when I save, then the system rejects it with an error.
- Given several milestones exist, when I view the project, then they are listed in date order.

US-04 Update milestone progress

- Given a milestone exists, when the engineer enters a completion value between 0 and 100 and saves, then the progress and update date are stored.
- Given the value is above 100 or negative, when I save, then it is rejected.
- Given progress is set to 100, when I save, then the status becomes "Completed".

US-06 Record material stock

- Given I am a store manager, when I add a material with name, quantity (>= 0), unit, supplier and reorder level, then it appears in the inventory list.

- Given I search by material name, when I submit, then matching results display in under 2 seconds.
- Given a negative quantity is entered, when I save, then it is rejected.

US-09 Register and assign workers

- Given I am a project manager, when I register a worker with name, skill and contractor, then a unique worker ID is generated.
- Given a worker exists, when I assign them to a project, then they appear in that project's workforce list.
- Given a worker is already assigned to the same project, when I assign again, then the system blocks the duplicate.

3.2 Remaining Stories (key criteria)

ID	Acceptance Criteria (Given-When-Then)
US-05	Given a milestone's planned end date has passed, when its progress is below 100%, then it is flagged "Delayed" and the manager is notified within 5 minutes.
US-07	Given stock falls below the reorder level, when an issue transaction is saved, then a low-stock alert appears on the dashboard and by email.
US-08	Given stock is 100 units, when 30 are issued to a milestone, then stock becomes 70 and the transaction is logged; issuing more than available is blocked.
US-10	Given a worker is assigned, when the supervisor marks attendance for a date, then it is saved once per worker per day and total hours are summarised.
US-11	Given I enter category, amount (> 0), date and description, when I save, then the expense is added and project total spend is updated immediately.
US-12	Given budget and expenses exist, when I open the budget view, then budget, actual, variance and % used show, and a warning appears above 90% of budget.
US-13	Given I log a risk with title, severity (Low/Medium/High) and owner, when I save, then it is added to the register with status "Open".
US-14	Given an open risk, when I change its status and add a mitigation note, then the change and timestamp are stored in its history.
US-15	Given project data exists, when I open the dashboard, then overall progress, delayed milestones, low-stock items, spend vs budget and open risks load in under 3 seconds.

US-16	Given I choose a project and date range, when I click Generate, then a PDF report is created in under 30 seconds with milestones, materials, costs and risks.

4. Prioritised Product Backlog

Method: MoSCoW combined with business value (1-10) and dependencies. Stories that other stories depend on (login, project, milestones) are ranked first, because nothing else can work without them.

Rank	ID	Story	Business Value	MoSCoW	Depends On	Reason
1	US-01	User registration and roles	9	Must	None	Security foundation
2	US-02	Create project	10	Must	US-01	Everything belongs to a project
3	US-03	Define milestones	10	Must	US-02	Core objective: milestone tracking
4	US-04	Update milestone progress	10	Must	US-03	Provides live progress data
5	US-06	Record material stock	8	Must	US-02	Material visibility
6	US-09	Register and assign workers	8	Must	US-02	Workforce records
7	US-11	Record expenses	9	Must	US-02	Cost control
8	US-05	Delay alerts	9	Must	US-03, US-04	Early identification of delays
9	US-07	Low-stock alert	7	Should	US-06	Prevents work stoppage
10	US-13	Log risks and issues	7	Should	US-02	Risk management

11	US-15	Dashboard	9	Should	US-04, US-06, US-11	Needs data from earlier stories
12	US-10	Attendance	6	Should	US-09	Productivity tracking
13	US-12	Budget vs actual	8	Should	US-11	Cost overrun control
14	US-08	Material usage	6	Could	US-06, US-03	Traceability
15	US-14	Risk status tracking	5	Could	US-13	Extends risk register
16	**US-16**	**Progress reports**	**8**	**Should**	**US-15**	**Needs all data in place**

5. Sprint 1 Backlog and Task Breakdown

Sprint length: 2 weeks (10 working days). Team: 5 members (2 backend, 1 frontend, 1 QA, 1 full-stack). Planned velocity: 24 story points.
Selected stories: US-01, US-02, US-03, US-04, US-06, US-09 (the top six ranks, all Must-haves forming the foundation).

Story	**Task**	**Estimate (hrs)**
US-01	Design users and roles tables	4
US-01	Build login and registration API with password hashing	8
US-01	Build login/register screens	6
US-01	Role-based access check and testing	6
US-02	Design project table and API (create, view, list)	5
US-02	Build project form with date and budget validation	5
US-02	Test project creation	2
US-03	Design milestone table and API	6
US-03	Build milestone form and list view	8
US-03	Validate dates against project dates, test	4
US-04	Build progress update API	4
US-04	Build progress input UI with progress bar	4
US-04	Auto-status logic and tests	3
US-06	Design material table and API	6

US-06	Build inventory form, list and search	8
US-06	Validation and testing	3
US-09	Design worker and assignment tables	4
US-09	Build worker registration and assignment UI	6
US-09	Duplicate-assignment check and testing	3
All	Integration testing, bug fixing, sprint demo prep	12
	Total	about 107 hrs

Capacity check: 5 members x 8 hrs x 10 days = 400 hrs, with about 50% allowed for meetings, learning, setup and reviews, giving about 200 productive hours. The 107-hour plan fits comfortably, leaving a buffer for rework and unknowns.

6. Story Point Estimation

Technique: Planning Poker using the Fibonacci scale (1, 2, 3, 5, 8, 13). Team members estimated privately, revealed together, and discussed differences until they agreed. US-02 (Create project) was the reference story at 3 points because it is a standard form with validation and a database table.

Story	Points	Justification
US-01 User registration and roles	5	Security work: password hashing, authentication and role checks add complexity and risk
US-02 Create project	3	Standard CRUD with simple validation; reference story
US-03 Define milestones	5	CRUD plus date rules against the parent project and sorted display
US-04 Update progress	3	Small update feature with a simple status rule
US-06 Record material stock	5	CRUD plus search and validation, more fields and screens than US-02
US-09 Register and assign workers	3	Two simple tables with one duplicate check
Total	24	Matches planned velocity

Stories that were *not* selected are sized for later planning: US-05 = 5, US-07 = 3, US-08 = 5, US-10 = 5, US-11 = 5, US-12 = 8, US-13 = 3, US-14 = 5, US-15 = 8, US-16 = 8. Large stories (8 points) will be split if needed before they enter a sprint.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

"By the end of Sprint 1, a project manager can securely log in, create a construction project, define milestones and update their progress, and the team can record materials and workers for that project."

7.2 Expected Deliverables

1. Secure login and role-based access for Manager, Engineer, Finance, Store and Admin users.
2. Project creation and listing module.
3. Milestone planning module with progress updates and automatic status.
4. Material inventory module with search.
5. Worker registration and project assignment module.
6. Tested, working increment deployed on a test server, with a database schema document, test report and sprint demo.

7.3 Definition of Done

- Code is reviewed and merged to the main branch.
- All acceptance criteria pass in testing.
- No open critical or high-severity bugs.
- Unit and integration tests pass.
- Feature is demonstrated to the Product Owner and accepted.

7.4 Sprint Risks and Mitigation

Risk	**Mitigation**
Security work (US-01) takes longer than estimated	Use a proven authentication library; start it on day 1
Unclear requirements from site staff	Short daily clarification with the Product Owner
Team member unavailable	Pair programming and shared task ownership

7.5 Scrum Ceremonies

- Sprint Planning: 2 hours on day 1.
- Daily Scrum: 15 minutes every day.
- Sprint Review: demo of the six stories on day 10.
- Sprint Retrospective: 1 hour after the review.

7.6 Next Sprints (Roadmap)

- Sprint 2: US-11, US-05, US-07, US-13 (expenses, delay alerts, low-stock alerts, risk log).
- Sprint 3: US-15, US-10, US-12, US-08 (dashboard, attendance, budget vs actual, material usage).
- Sprint 4: US-14, US-16, hardening, performance and user acceptance testing.